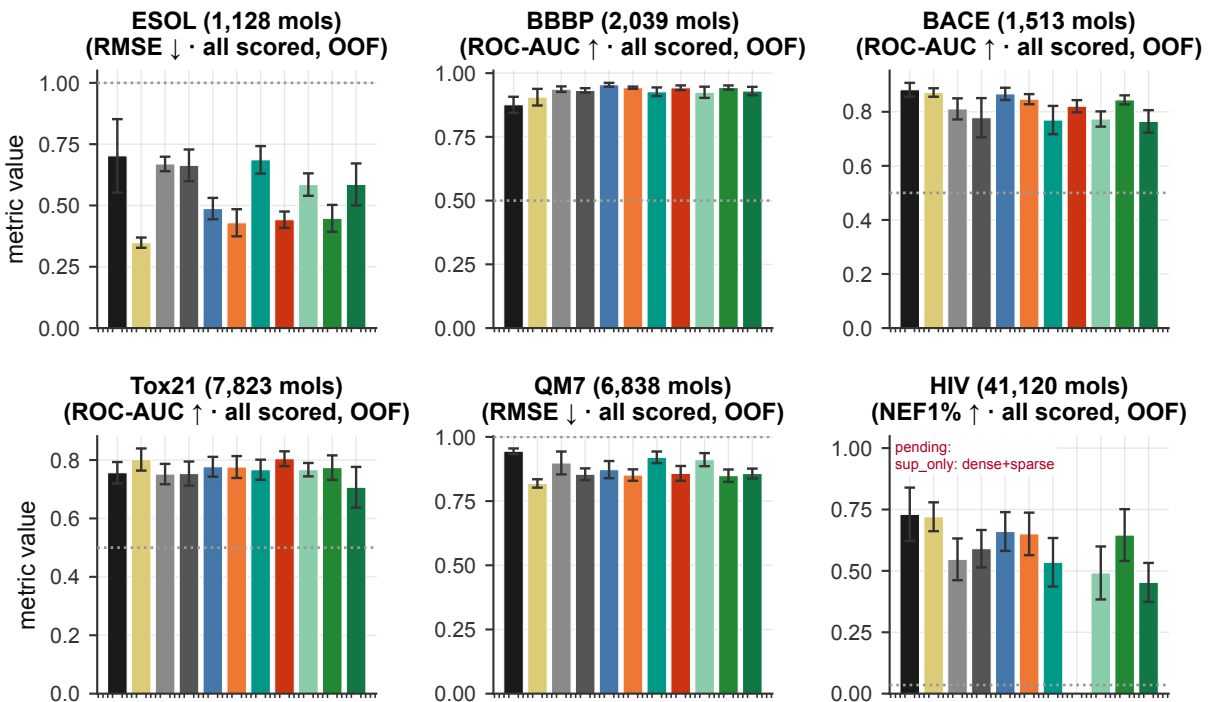


Fig A1.b - which model performs best? (matched at 8M forward passes)

pooled 5-fold scaffold cross-validation (README §8.1 protocol of record) · error bars = ± 1 sd across the 5 scaffold folds
 · classical anchors: 3 XGBoost seeds · no_pretrain (frozen) = random-init encoder, features frozen; no_pretrain (end-to-end) = same encoder fine-tuned on the task



Morgan+XGBoost (ECFP4)
 no_pretrain (end-to-end)
 sup_only: sparse_all
 unsup→sup: dense (MTR)

Morgan+desc+XGBoost
 unsup_only (MLM)
 sup_only: dense+sparse
 unsup→sup: dense+sparse

no_pretrain (frozen)
 sup_only: dense (MTR)
 unsup→sup: sparse_all
 random (chance)